

How Networks Are Used ?

- **Network Services:** Resources a network provides to its users.
Includes applications and their provided data.

Example: When you access a website or use cloud storage, the network is providing you with services like **data retrieval, file sharing, and communication**.

- **Common Network Applications:**
 - **Client-Server Model** !
 - **File and Print Services**
 - **Communication Services**

1- Client-Server Applications

- A **client** requests data or services from a **server**. A server stores and manages data, while clients (computers, phones, etc.) request and access that data.
- **Common client-server applications:**
 - **Web service** → Accessing websites
 - **Email services** → Sending and receiving emails
 - **FTP service** → Transferring files
 - **Telnet service** → Remote command-line access
 - **Remote Desktop** → Controlling another computer
 - **Remote applications** → Running apps on a server

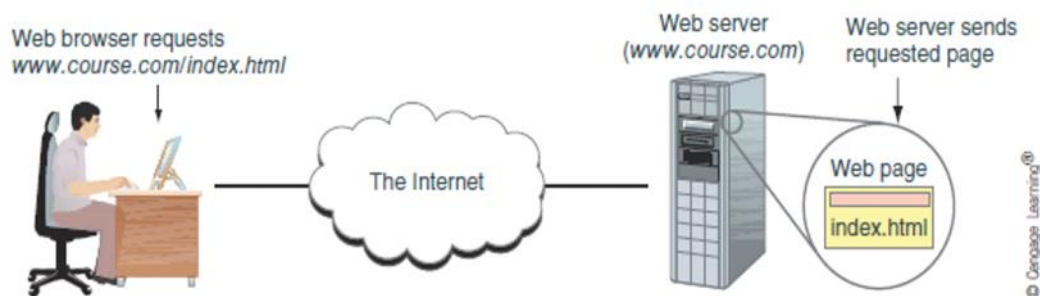


Figure 1-1 A Web browser (client application) requests a Web page from a Web server (server application); the Web server returns the requested data to the client

- A **web browser (client)** requests a webpage from a **web server** by entering a URL (e.g., www.course.com/index.html).
- The request travels through **the internet** to reach the **web server**.
- The **web server** processes the request and sends back the requested **webpage (index.html)**.
- The **browser** then displays the webpage to the user.

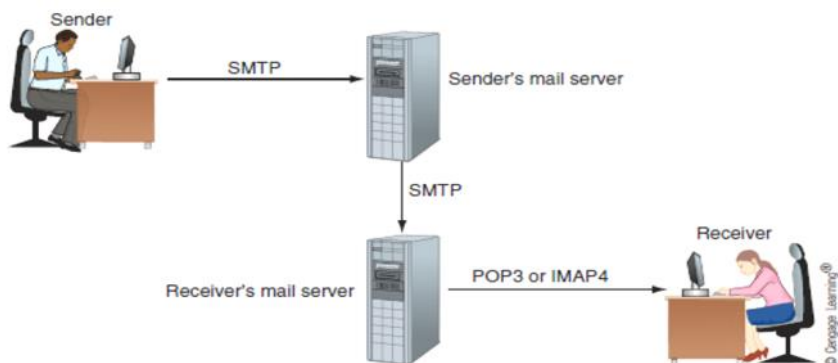


Figure 1-2 SMTP is used to send email to a recipient's mail server, and POP3 or IMAP4 is used by the client to receive email

How Email Works in a Network:

1. **Sender (Client)** writes an email and sends it using **SMTP (Simple Mail Transfer Protocol)**.
2. The email is first transmitted to the **Sender's Mail Server**.
3. The sender's mail server uses **SMTP** again to forward the email to the **Receiver's Mail Server**.
4. The **Receiver retrieves the email** using either:
 - **POP3 (Post Office Protocol 3)** – Downloads and removes the email from the server.
 - **IMAP4 (Internet Message Access Protocol 4)** – Allows the receiver to view the email while keeping it on the server.

2- File and Print Services

! (learn meaning of file and print service)

- **File Services:** A server's ability to **share files and storage**.
- **File Server:** A computer that provides file-sharing services.
- **Print Services:** Allows multiple users to share a printer over a network.
 - Reduces maintenance and management effort with a single printer.

3- Communication Services

! (learn communication ser. and types)

Convergence = kümelenme

- **Convergence:** Using a single network for multiple communication services.
- **Unified Communication (UC):** Centralized management of network-based communications.
- **Types of Communication Services:**
 - **VoIP (Voice over IP):** Internet-based voice communication.
 - **Live Streaming:** Real-time audio and video transmission.
 - **Stored Streaming:** Playback of pre-recorded audio and video.
- **Voice and video need low delay** for smooth transmission.
- **They can tolerate some data loss** without major issues.
- **Network admins must ensure good QoS (quality of service)** for voice and video.
- **Bandwidth affects network traffic and data flow.**

Controlling Network Access

! (physical and logical topology)

- **Topology:** How network components work together.
- **Physical Topology:** It explains how computers, devices (routers, switches, modems, etc.), and cables are arranged and connected in a network.
- **Logical Topology:** How access to the network is managed **via** software.
- **Network Operating System:** Controls network access, essential for client-server models.

! Peer-to-Peer (P2P) Model ! (P2P and client server differences)

- **P2P network model** means that each computer manages its **own** resources and **access** permissions.
- **No centralized control:** There is no main server; every device (computer) acts as both a client and a server.
- **Computers in the network are called nodes or hosts:**
 - They form a logical group where each computer can decide to **share resources (files, printers, internet, etc.)** or **restrict access** to them.
- **Each user has a Windows local account:**
 - The user can only log in and access data on their own computer, not on other devices in the network.

This model is **simple to set up** and works well for **small networks** (e.g., home or small office networks), but it **lacks centralized security and control**.

CHAPTER 1 Introduction to Networking

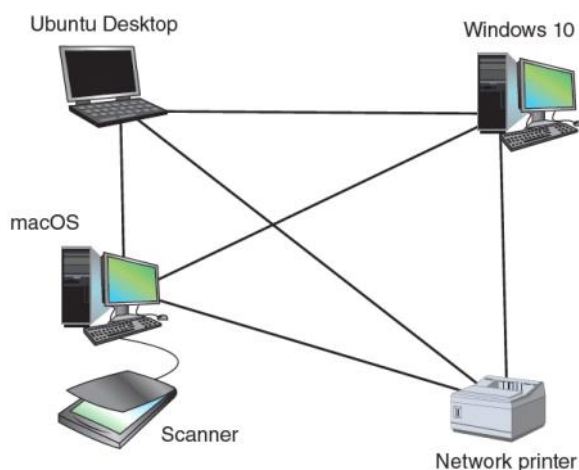


Figure 1-1 In a peer-to-peer network, no computer has more authority than another; each computer controls its own resources, and communicates directly with other computers

The lines in the diagrams show the **logical topology**, meaning how computers **communicate** with each other, not how they are **physically placed**.

Advantages

- **Simple Configuration:** Easy to set up without requiring special hardware or software.
- **Cheaper:** Cheaper than other network models since no central server is needed.
- **Good for small business**

Disadvantages

- **Not Scalable:** Difficult to expand as more devices are added.
- **Not Secure:** Lacks centralized security, making it vulnerable to unauthorized access.
- **Not Practical for Large Installations:** Works well for small networks but inefficient for large businesses.

Client-Server Network Model

- In the **client-server model**, a special system called the **Network Operating System (NOS)** manages **all** computers and their resources using a **centralized directory database**.
- A group of computers is managed by **Windows Server** is called a **Windows domain**.
- The centralized directory database that contains **user account information** and security for the entire group of computers is called **AD (Active Directory)**.
- Each user gets a **global account** (also called a **network ID**) from the **network administrator**. This account is saved in **Active Directory** and lets users access the network and its resources
- In a **client-server network**, users can log in from any computer in the network. Their **access to files and services** is controlled by **Active Directory (AD)**.
- This process is managed by **Active Directory Domain Services (AD DS)**.
- **Clients do not share resources directly** with each other. Instead, they **request** access through a **centralized directory database**, which decides what they can use.

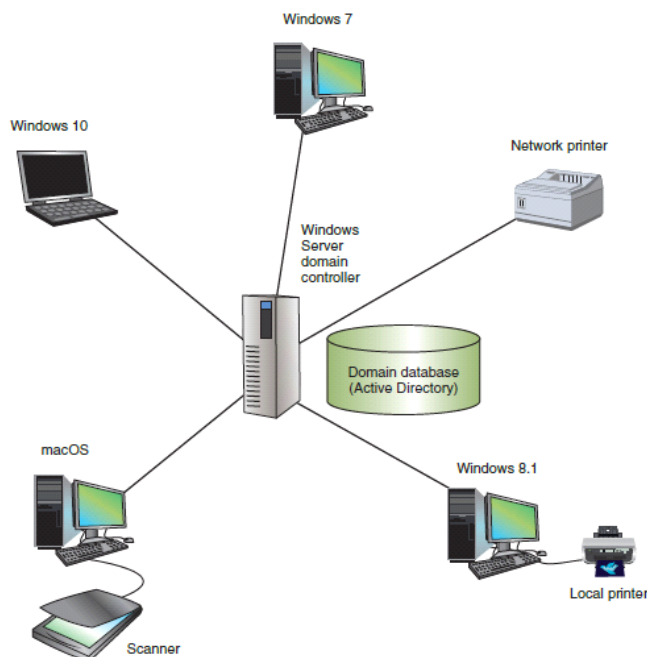


Figure 1-2 A Windows domain uses the client-server model to control access to the network, where security on each computer or device is controlled by a centralized database on a domain controller

the **NOS** (for example, Windows Server 2016, Ubuntu Server, or Red Hat Enterprise Linux) is **responsible for:**

- Manages client data and other resources
- Ensures only authorized user access the network.
- It **controls** which files users can **open and read**.
- Restricting when and from where users can access the network
- Dictates computer communication rules
- Supplies application to clients

Servers with **NOS** need more **memory**, processing power and storage capacity than clients. because they handle many users at once.(loads and request from multiple clients)

Client-server networks are **more complex** than peer-to-peer networks, but they have many **advantages**:

- **User accounts and passwords** are assigned from one place.
- Users can get access to **shared files or printers** easily.
- **Problems** can be monitored, diagnosed and fixed from one central location.
- It is **easy to add new devices**, so the network can grow (scalable).

These networks are good for **bigger systems** and are controlled by a **Network Operating System (NOS)**.

Network Hardware and Physical Topologies

A **Local Area Network (LAN)** is a small network located in a limited area like an office or building.

It includes some hardware components such as:

- A **switch**, which receives data from one device and forwards it to the correct destination port.
- A **Network Interface Card (NIC)**, which allows a device to connect to the network. !
- In a **star topology**, all devices are connected to a central device (typically a switch), improving communication efficiency. where each device connects to multiple other devices, which is called a **mesh topology**.

(NIC meaning and functions of them)

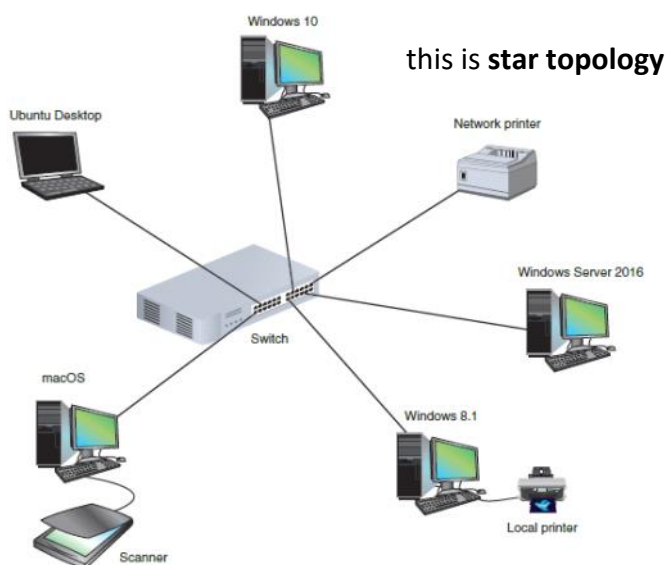


Figure 1-6 This LAN has five computers, a network printer, a local printer, a scanner, and a switch, and is using a star topology

✕ Hybrid Topology

- A network that combines **different topology types** is called a **hybrid topology**.
- It provides **greater flexibility** and is easier to **scale**.

🔌 LAN and Switches

- A **LAN** can include **multiple switches** to connect more devices.
- These switches improve **network performance and coverage**.

🔗 Backbone (network of networks)

- A **backbone** is a **central conduit** that links different parts (segments) of the network.
- It usually offers **higher transmission speeds** and may use **special cabling**.

🔄 Bus Topology

- When **multiple switches are connected in a line**, this forms a **bus topology** (daisy-chained).

★ Star-Bus Topology

- If each switch in the bus topology connects to devices in a **star layout**, it becomes a **star-bus topology**.

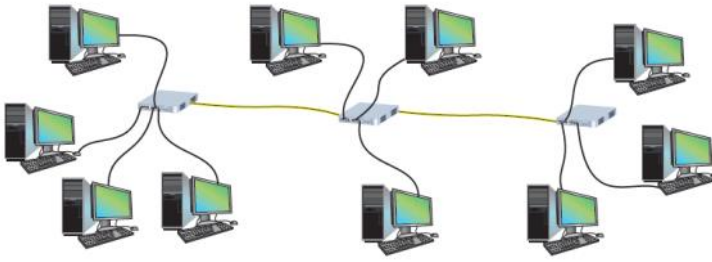
star-bus topology

Figure 1-10 This local network has three switches, and is using a star-bus topology

ring topology

Figure 1-11 Using a ring topology, a computer connects to the two computers adjacent to it in the ring

- A **router** is a device that directs network traffic between **two or more networks** and helps determine the **best path** for data to travel.
- In **home networks**, routers are used to connect the **LAN to the Internet**.
- **Industrial routers** often have **multiple ports**, allowing them to connect to several networks at once.
- The **main difference** between a **router** and a **switch** is their function:
 - A **switch** connects devices within the same network.
 - A **router** acts as a **gateway** between different networks.

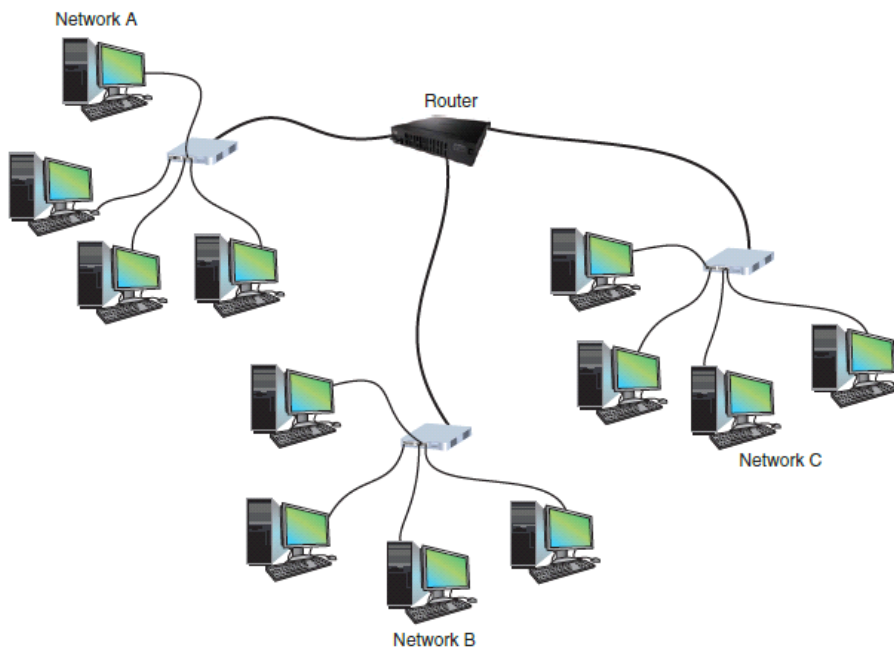


Figure 1-13 Three LANs connected by a router

There are 3 types of **computer network** ;

- LAN
- MAN
- WAN

MANs and WANs

- A **MAN (Metropolitan Area Network)** connects multiple **LANs** within a single city or campus. It is sometimes called a **Campus Area Network (CAN)**.
- A **WAN (Wide Area Network)** connects LANs across **large geographical areas**. The **Internet** is the biggest example of a WAN.
- **MANs and WANs** often use **different technologies** and **transmission media** compared to LANs.
- A **PAN (Personal Area Network)** is the **smallest type of network**, used for personal devices like phones, tablets, or laptops.

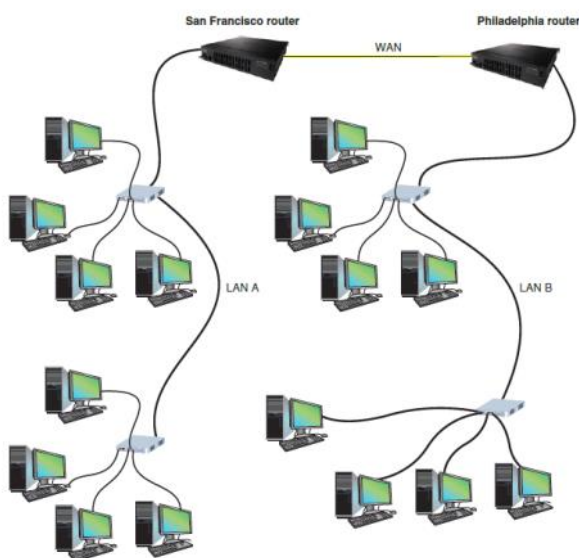
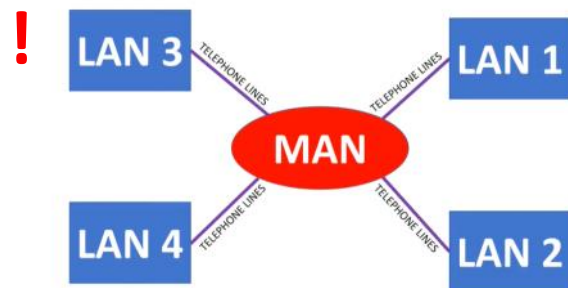


Figure 1-14 A WAN connects two LANs in different geographical areas



The Seven-Layer OSI Model

The **OSI (Open Systems Interconnection)** model is a **seven-layer framework** created to describe how communication happens in a network.

The seven layers are:

1. **Physical**
2. **Data Link**
3. **Network**
4. **Transport**
5. **Session**
6. **Presentation**
7. **Application**

Each layer has a specific role in ensuring data is transmitted and understood correctly between devices.

! (he may ask one of the layer definitions(not all of them))

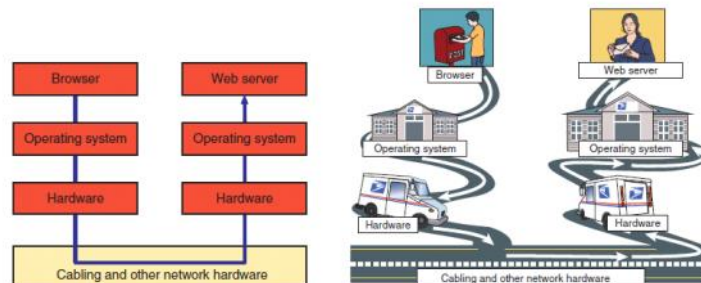


Figure 1-15 A browser and web server communicate by way of the operating system and hardware, similar to how a letter is sent through the mail using the U.S. Postal Service and the road system

The Seven Layer of OSI Model

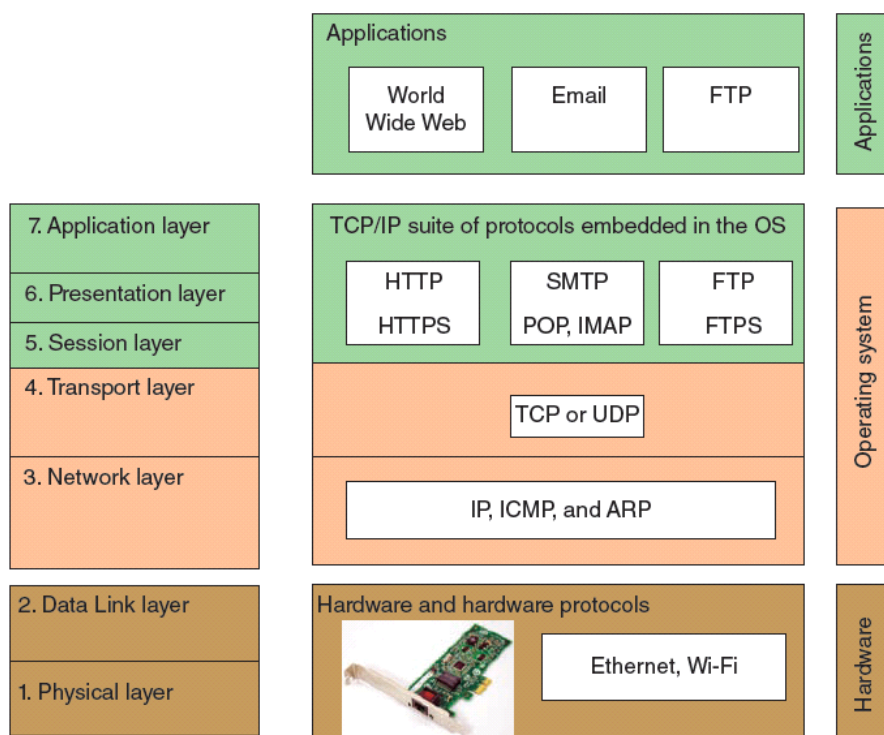


Figure 1-16 How software, protocols, and hardware map to the seven-layer OSI model

Layer 7: Application Layer

The **Application Layer** is the top layer of the OSI model. It manages communication between two applications running on different computers.

There are two main types of **application layer protocols**:

- **Provide services to a user**, such as **web browsers** and web servers.
- **Utility protocols**, like **SNMP** (Simple Network Management Protocol), help the system by **monitoring network traffic** and **collecting data**. They can also **send alerts** to network admins when something goes wrong or needs attention.

The **payload** is the actual **data exchanged between applications** or services and the operating system. This layer ensures that data is ready for users or programs to work with.

Layer 6: Presentation Layer

The **Presentation Layer** prepares data so that the **receiving application** can understand it.

It is responsible for:

- **Reformatting** the "data"
- **Compressing** it to reduce size
- **Encrypting** it to keep it secure

 **Example:** An email message can be **encrypted** at this layer by the **email client** or the **operating system**.

Layer 5: Session Layer

The **Session Layer** manages how **data is synchronized and recovered** between applications, especially if some parts are lost or damaged during transfer.

It works closely with the **Application** and **Presentation** layers, and it's often hard to tell them apart because their functions overlap.

Most actions in this layer happen when an application uses an **API (Application Programming Interface)** to send a request to the **operating system (OS)**.

Layer 4: Transport Layer

The **Transport Layer** moves **application data (payloads)** from one device to another and ensures correct delivery.

There are two main protocols used in this layer:

- **TCP (Transmission Control Protocol)**
 - Creates a **reliable connection** before sending data. (connection w end host)
 - Checks if the data was delivered successfully.
 - Known as a **connection-oriented protocol**.
- **UDP (User Datagram Protocol)**
 - Sends data **without establishing a connection**.
 - Does **not confirm** if the data was received.
 - Known as a **connectionless protocol**.

At the Transport Layer, protocols add a **header** at the beginning of the data (payload). This header contains **control information**. // data basına header eklenir. buna encapsulation denir. dataların gideceği yerler karışmasın diye.

This process is called **encapsulation**—it means adding a header to data received from the layer above.

The **header** includes a **port number**, which helps the message reach the correct **application** on the receiving device.

If the message is **too large**, **TCP** splits it into smaller pieces called **segments**.

With **UDP**, these pieces are called **datagrams**.

Layer 3: Network Layer

The **Network Layer** is responsible for moving messages from one device (node) to another until they reach the final destination.

- The **IP protocol** adds a **Network Layer header** to the data, creating a full message called a **packet**.
- Each device on a network has a unique **IP address**, which helps identify where the packet should go.
- The Network Layer uses **routing protocols** (like **ICMP** and **ARP**) to find the **best path** for the packet to travel across networks.

Firmware refers to programs embedded into hardware devices. This software does not change unless a firmware upgrade is performed.

Layer 2: Data Link Layer

The **Data Link Layer** (Layer 2) works with **Layer 1** to connect devices to the **local network hardware** like network cards (NICs).

- The protocols used here are usually **built into the device firmware**.
- The type of connection (e.g., **Ethernet** or **Wi-Fi**) determines which protocol is used.
- This layer adds a **header and trailer** around data to control how it's sent—this full data unit is called a **frame**.

It also uses **MAC addresses (Media Access Control)** to identify the **source and destination** devices.

- MAC addresses are also called **physical addresses** and are built into every network adapter.
- They are used to find devices **only within the local network** (short-range communication).

Layer 1: Physical Layer

The **Physical Layer** is the **lowest and simplest layer** of the OSI model. It is responsible for the **actual transmission of bits** (0s and 1s) over a **wired or wireless** connection.

Data can be sent in different physical forms, such as:

- **Waves** in the air (for wireless signals),
- **Electric voltage** through copper wires,
- Or **light signals** using fiber-optic cables.

This layer focuses only on how raw data is physically moved—not on its meaning.

A **Protocol Data Unit (PDU)** is the technical term for a group of bits that is passed from one OSI layer to another, or between networks. While technicians may casually call it a “message” or “transmission,” each OSI layer has a specific name for this unit.

Table 1-1 Names for a PDU or message as it moves from one layer to another		
OSI model	Name	Extremely technical name
Layer 7, Application layer Layer 6, Presentation layer Layer 5, Session layer	Payload or data	L7PDU
Layer 4, Transport layer	Segment (TCP) or datagram (UDP)	L4PDU
Layer 3, Network layer	Packet	L3PDU
Layer 2, Data Link layer	Frame	L2PDU
Layer 1, Physical layer	Bit or transmission	L1PDU

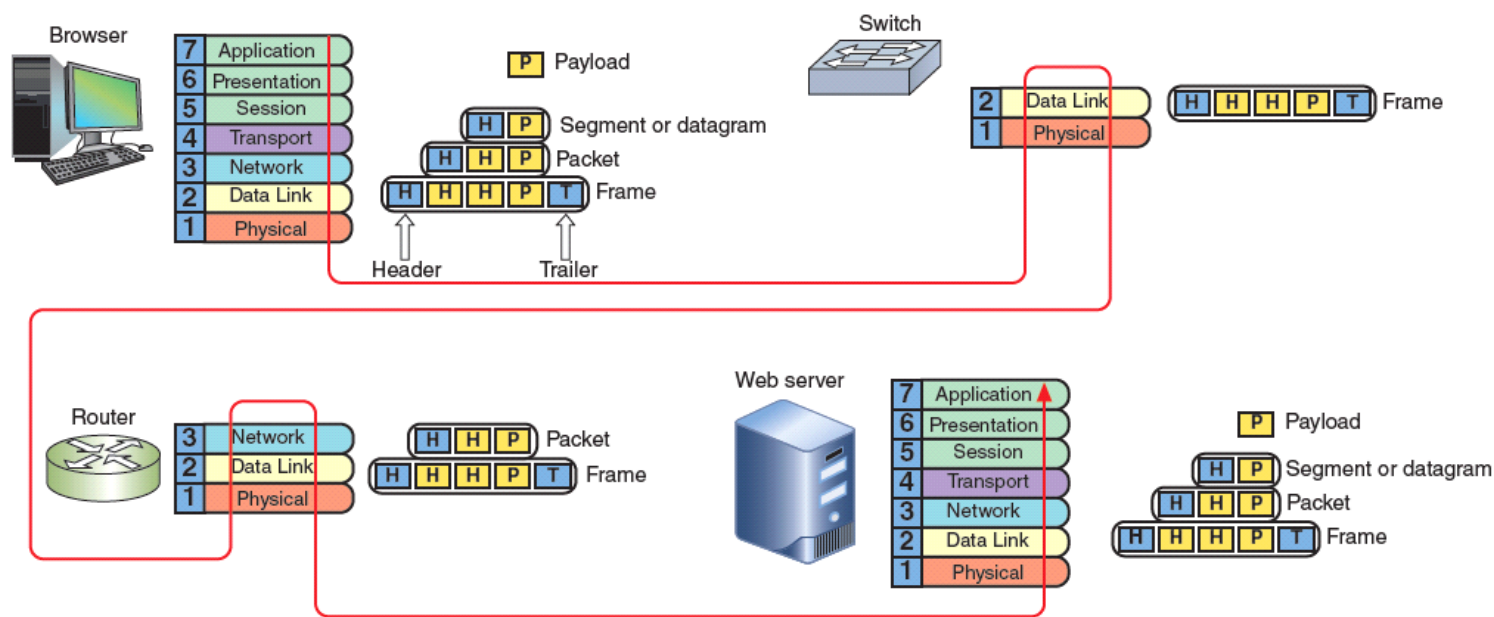


Figure 1-17 Follow the red line to see how the OSI layers work when a browser makes a request to a web server

Table 1-2 Steps through the OSI layers during a browser-to-web server transmission	
Sending host	1. The browser, involving the Application, Presentation, and Session layers, creates an HTTP message, or payload, on the source computer and passes it down to the Transport layer.
	2. The Transport layer (TCP, which is part of the OS) encapsulates the payload by adding its own header and passes the segment down to the Network layer.
	3. IP at the Network layer in the OS receives the segment (depicted as two yellow boxes in the figure), adds its header, and passes the packet down to the Data Link layer.
	4. The Data Link layer on the NIC firmware receives the packet (depicted as three yellow boxes in the figure), adds its header and trailer, and passes the frame to the Physical layer.
	5. The Physical layer on the NIC hardware puts bits on the network.
Switch	6. The network transmission is received by the switch, which passes the frame up to the Data Link layer (firmware on the switch), looks at the destination MAC address, and decides where to send the frame.
	7. The pass-through frame is sent to the correct port on the switch and on to the router.
Router	8. The router has two NICs, one for each of the two networks to which it belongs. The Physical layer of the first NIC receives the frame and passes it up to the Data Link layer (NIC firmware), which removes the frame header and trailer and passes the packet up to IP at the Network layer (firmware program or other software) on the router.
	9. This Network layer IP program looks at the destination IP address, determines the next node en route for the packet, and passes the packet back down to the Data Link layer on the second NIC. The Data Link layer adds a new frame header and trailer appropriate for this second NIC's LAN, including the MAC address of the next destination node. It passes the frame to its Physical layer (NIC hardware), which sends the bits on their way.
Destination host	10. When the frame reaches the destination host NIC, the Data Link layer NIC firmware receives it, removes the frame header and trailer, and passes the packet up to IP at the Network layer, which removes its header and passes the segment up to TCP at the Transport layer.
	11. TCP removes its header and passes the payload up to HTTP at the Application layer. HTTP presents the message to the web server.

// SUMMARY

- Networks provide a wide range of services, including client-server applications, file and print services, and communications services.
- File and print services enable multiple users to share data, storage areas, and printers.
- Traditional peer-to-peer networks are usually simple and inexpensive to set up.
- The client-server model for access control relies on a centrally administered server using a NOS that manages shared resources for multiple clients.
 - More complex and expensive to install.